



School of Computer Applications and Technology
Mid Term Examination - Nov 2025

Masters of Computer Applications

Duration : 90 Minutes
Max Marks : 50

Sem I - E1PA104B - Database Management System

General Instructions

Answer to the specific question asked

Draw neat, labelled diagrams wherever necessary

Approved data hand books are allowed subject to verification by the Invigilator

- 1) How a database management system overcomes the issues of a file system to provide better data handling? K1 (3)
- 2) Define overall structure of a database system. K1 (3)
- 3) A School Database allows teachers to update marks, summarize, which database languages will be used? Explain with suitable example and syntax. K2 (4)
- 4) Explain three-level architecture of DBMS with a suitable diagram. K2 (4)
- 5) Identify the role of database schema in a retail store for efficiently organizing, storing and maintaining product and sales records. K3 (5)
- 6) Construct an E-R diagram for an e-learning platform that includes Students, Courses, and Instructors. K3 (5)
- 7) Analyze the three level schema architecture. Using a example to show how logical data independence and physical data independence are achieved. K4 (8)
- 8) Analyze an E-R diagram to determine the primary key and foreign key relationships that will result after mapping. Provide suitable example. K4 (8)
- 9) A bank processes multiple customer transactions such as fund transfers and loan approvals while ensuring that only authorized staff can modify sensitive financial data. To maintain data security and consistency, the bank uses DCL commands to control access and TCL commands to manage transaction integrity. Explain the justification for the given scenario. K5 (10)

OR

Justify with suitable, E-R diagram for a Hospital Management System using multiple relationship types. Mention the reasoning behind each relationship type you used. K5 (10)